B.TECH CAPSTONE THESIS PRESENTATION | FINAL TECHNICAL EVALUATION

DESIGN AND IMPLEMENTATION OF AN INTERNET-ENABLED ACOUSTIC LEVITATION MANIPULATION SYSTEM

ESP32 APB Clock Synchronization, Dual Push-Pull Amplification, and Real-Time IoT Teleoperation

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PROJECT INVESTIGATORS:

- RONIL (24BCE1330) — B.Tech CSE
- TEJAS (24BCE1360) — B.Tech CSE
- PRIYANSHU (24BCE1140) — B.Tech CSE
- PRASHANT (24BCE1118) — B.Tech CSE

FACULTY GUIDANCE & LEADERSHIP:

Project Guide: Dr. Mohammed Aarif K.O., Professor, SCOPE
Dean: Dr. Viswanathan V, Professor & Dean, SCOPE
Associate Dean: Dr. Suganya G, Professor & Associate Dean, SCOPE
School: School of Computer Science and Engineering (SCOPE)
Institution: Vellore Institute of Technology (VIT), Chennai – Nov 2026

EXECUTIVE SUMMARY

The 45-60 Second Presentation Pitch

High-Impact Rapid Project Overview and Core Value Proposition

TIMED SPEAKER NARRATIVE (45–60 SECONDS):

[00–10s] The Challenge: Handling fragile micro-objects, living cells, and pure chemical reagents without solid surface friction, electrostatic distortion, or petri-dish contamination has historically required $5,000+ laboratory setups.

[10–25s] The Solution: We engineered an autonomous, internet-enabled acoustic levitator using 6 piezoelectric transducers and an ESP32 microcontroller, creating deep ultrasonic potential wells that trap and levitate objects in mid-air.

[25–45s] The Innovation: We eliminated bulky transformers by inventing a dual L293D push-pull driver delivering 24 Vpp from a 12V supply. We synchronized the ESP32 80 MHz APB clock to achieve 1.4° phase steps, enabling sub-millimeter vertical manipulation (~12 µm/deg).

[45–60s] The Impact: Featuring native Wi-Fi REST web teleoperation and a total bill-of-materials under $28 at only 5.6 Watts, this system democratizes contactless handling for cleanrooms, bio-assays, and micro-assembly.

KEY METRIC CALLOUTS:

154.2 dB
Peak Cavity Acoustic SPL

24 Vpp
Differential Drive (No Transformers)

1.406°
Hardware Timer Phase Granularity

< $28
Total System Bill of Materials

5.6 W
Ultra-Low Power Consumption

< 15 ms
Embedded Wi-Fi REST Latency

SOLUTION OVERVIEW

End-to-End System Integration & Architecture

Harmonizing High-Speed Embedded Firmware, Push-Pull Power Drivers, and Acoustic Physics

FOUR-TIER INTEGRATED ARCHITECTURE:

1. High-Speed Compute Core (ESP32):

Dual-core 240 MHz Tensilica LX6. Hardware APB timer (LEDC) generates 4 complementary square waves at 40 kHz with zero clock jitter.

2. Transformerless Power Stage (Dual L293D):

Differential push-pull H-bridge topology inverts output polarities, boosting a single 12V rail to 24 Vpp directly across transducer terminals.

3. Resonant Cavity (6x 40 kHz PZT):

3 top and 3 bottom transducers tilted inward at 12° across a calibrated distance D = 72.25 mm (17 semi-wavelengths), superimposing acoustic waves to exceed 154 dB SPL.

4. Zero-Install Teleoperation (Wi-Fi Web UI):

FreeRTOS Core 1 hosts an asynchronous HTTP REST server and interactive web interface for sub-millimeter particle translation and diagnostics.

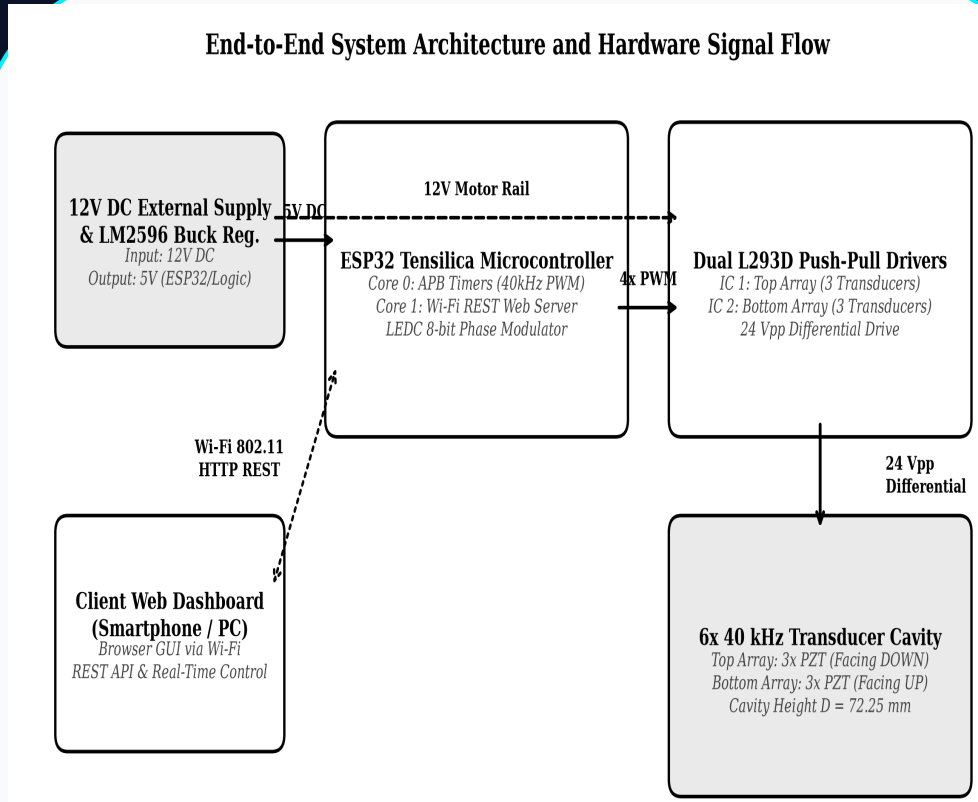


Figure 3.1: End-to-End System Block Diagram & Hardware Signal Flow

What the Hardware Does: Signal Generation & Push-Pull Drive

Generating 24 Vpp Differential Excitation from a 12V Supply without Step-Up Transformers

HARDWARE WAVEFORM & POWER STAGE:

Direct APB Clock Sourcing (80 MHz):

Hardware timer registers lock directly onto the 80 MHz Advanced Peripheral Bus, eliminating CPU-induced jitter and guaranteeing microsecond phase stability.

Differential Push-Pull Inversion (180°):

Rather than driving transducers against ground (0 to 12V), each transducer is driven across two opposing half-bridges: Phase A (0°) on terminal 1 and Phase B (180°) on terminal 2.

Transformerless Voltage Doubling (24 Vpp):

When Phase A is HIGH (12V) and B is LOW (0V), Vdiff = +12V. In the next half-cycle, polarities invert: Vdiff = -12V. Total swing = 24 V peak-to-peak.

Acoustic Power Quadrupling (+13.6 dB):

Acoustic radiation pressure scales with applied voltage ($P0 \propto Vpp$) and acoustic power scales quadratically ($Pac \propto Vpp^2$), boosting energy output 23x compared to 5V driving.

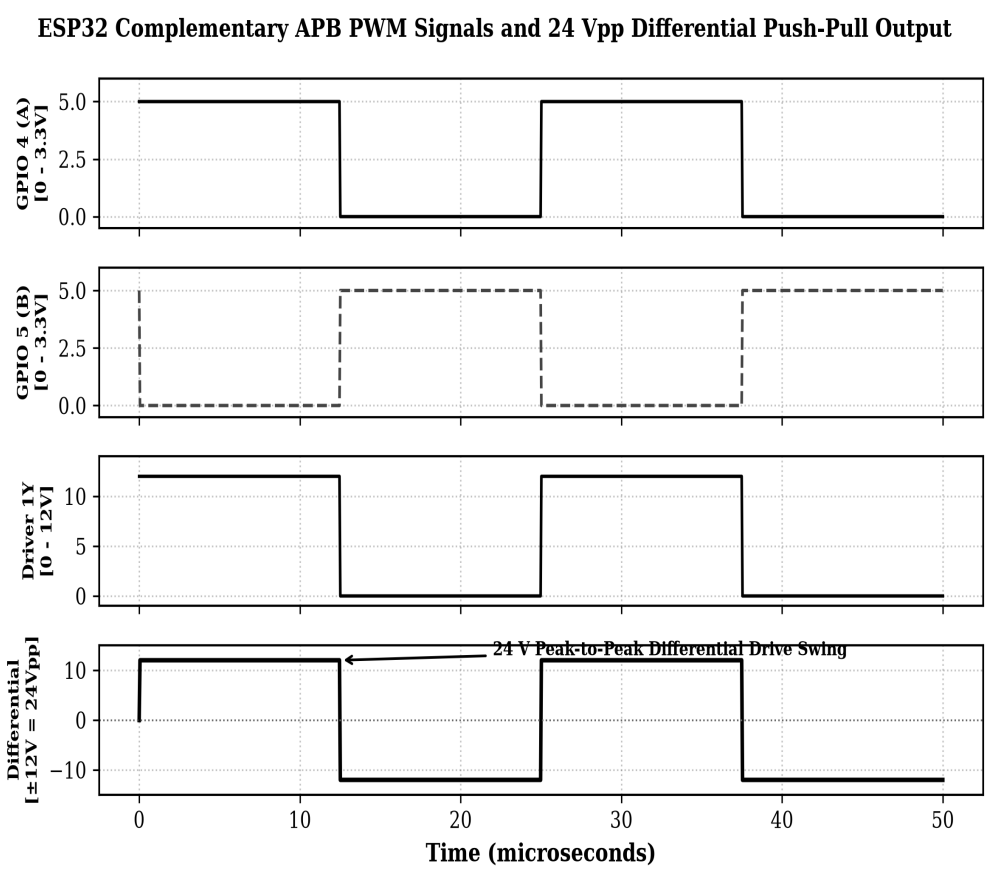


Figure 3.4: Complementary PWM Drive and 24 Vpp Differential Waveforms

HARDWARE ENGINE (PART 2)

What the Hardware Does: Cavity Geometry & Resonance

Optimized 6-Transducer Opposing Arrays and Multi-Node Standing Wave Physics

ACOUSTIC CAVITY & TRANSDUCER CLUSTERS:

6-Transducer Cluster Topology:

Six 16 mm 40 kHz piezoelectric transducers arranged as two opposing arrays: 3 top (facing DOWN) and 3 bottom (facing UP) in equilateral triangles (17.5 mm pitch).

12.0° Inward Concave Focal Tilt:

Each transducer is geometrically angled inward at 12° toward the central vertical axis, ensuring constructive acoustic wave superposition at the center trapping node.

Calibrated Resonant Height (D = 72.25 mm):

Cavity height enforces the acoustic semi-wavelength resonance condition: D = 17 × (λ / 2) ≈ 72.25 mm at 20°C, establishing 17 stationary trapping nodes in air.

Power Isolation & Filtering:

LM2596 buck regulator delivers isolated 5.0V logic power; dual 100 µF electrolytic and 0.1 µF ceramic bypass capacitors absorb 40 kHz inductive switching spikes.

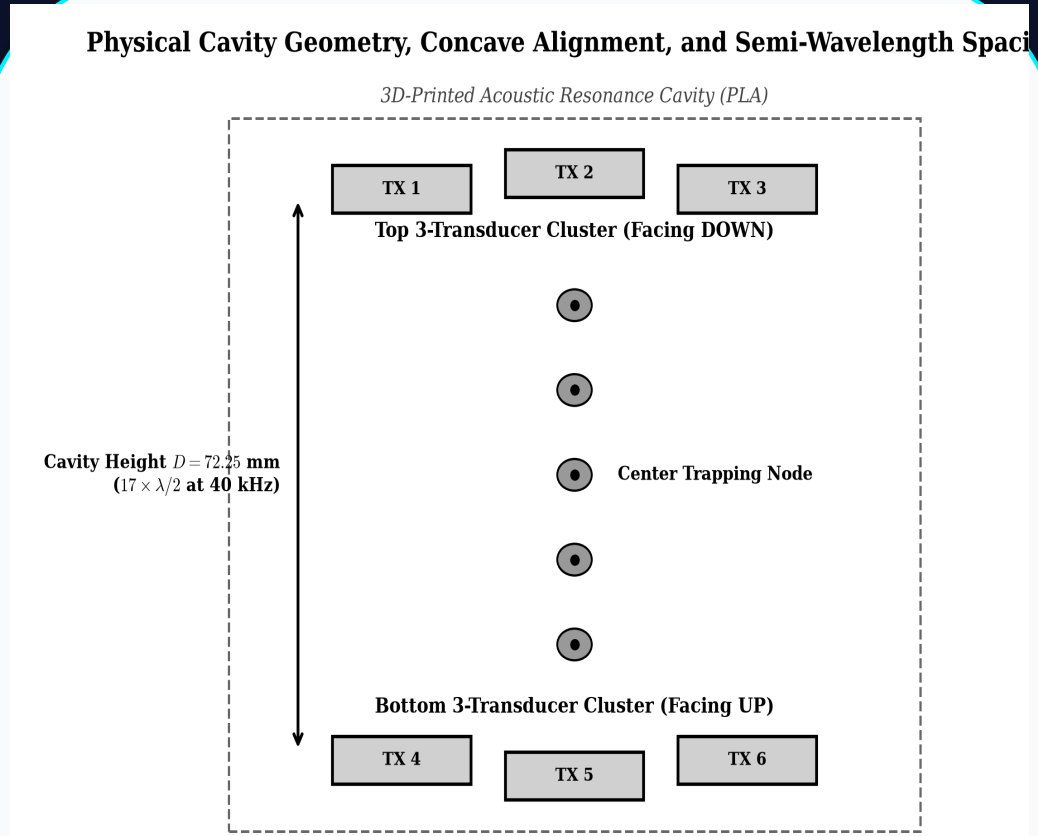


Figure 3.2: 6-Transducer Cavity Geometry, Concave Alignment, and Spacing

ELECTRICAL INTEGRATION

Master Circuit Schematic & Pin Interconnections

Deterministic GPIO Routing, Dual H-Bridge Drivers, and Star-Ground Isolation

HARDWARE PIN INTERCONNECTION MATRIX:

ESP32 GPIO 4: IC 1 Pins 2 & 15 (1A, 4A) -> Top Phase A (0° ref)

ESP32 GPIO 5: IC 1 Pins 7 & 10 (2A, 3A) -> Top Phase B (180° inv)

ESP32 GPIO 18: IC 2 Pins 2 & 15 (1A, 4A) -> Bot Phase A (Phase shift)

ESP32 GPIO 19: IC 2 Pins 7 & 10 (2A, 3A) -> Bot Phase B (180° push-pull)

5V Rail (LM2596): ESP32 VIN, IC 1 & 2 VCC1 (Pin 16), EN (Pins 1 & 9)

12V DC Input: IC 1 & IC 2 VCC2 (Pin 8) motor power rail

IC 1 Outputs: Pins 3 & 6 -> Top TX #1, #2 | Pins 14 & 11 -> Top TX #3

IC 2 Outputs: Pins 3 & 6 -> Bot TX #4, #5 | Pins 14 & 11 -> Bot TX #6

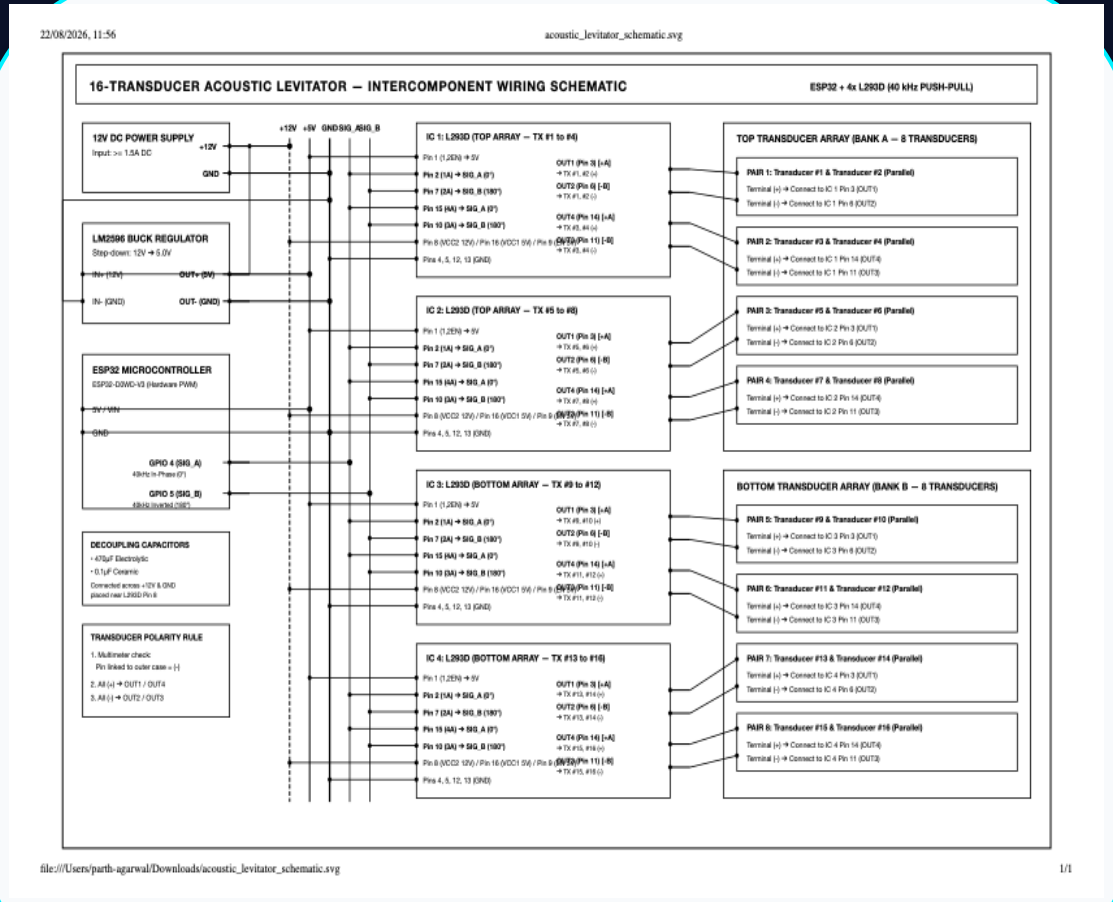


Figure A2.1: Master Electrical Circuit Schematic & Wiring Layout

SOFTWARE ENGINE (PART 1)

What the Software Does: APB Timer Registers & Core Isolation

Hardware-Synchronized Zero-Jitter PWM Synthesis and Dual-Core FreeRTOS Partitioning

LEDC Hardware Timers

Direct Register Control:

Configures LEDC_TIMER_0 with high-speed 80 MHz APB clock routing, bypassing software interrupts entirely.

8-Bit Counter (256 Ticks):

Each 25 µs period is segmented into 256 discrete ticks, enabling fine 1.406° phase resolution per tick.

Exact Fractional Divider:

Prescaler set to 7 + 832/1024, synthesizing exactly 40,000.0 Hz with zero cumulative clock drift.

Dual-Core FreeRTOS Tasks

Core 0 (Deterministic PWM):

Dedicated to real-time timer comparator registers, GPIO phase modulation, and 115200 baud USB serial command CLI.

Core 1 (Asynchronous IoT):

Runs the 2.4 GHz 802.11n Wi-Fi radio stack, TCP/IP socket buffer, and embedded RESTful HTTP web server.

Zero Phase Jitter:

Network traffic and HTTP requests on Core 1 never introduce latency or jitter into the ultrasonic wave generation on Core 0.

Dynamic Frequency Trimming

Real-Time Resonant Tuning:

Allows ±100 Hz frequency adjustment on the fly (38.5 kHz to 41.5 kHz) via web or serial command.

Thermal Drift Tracking:

Compensates for transducer piezoelectric warming during extended levitation without rebooting the controller.

Automated Frequency Sweep:

Sweeps 38.0 kHz to 42.0 kHz to automatically identify cavity acoustic standing wave resonance.

SOFTWARE ENGINE (PART 2)

What the Software Does: Dynamic Phase Shifting & Manipulation

Linear Axial Particle Translation via On-The-Fly Hardware Timer Offset Registers

PHASE-TO-DISPLACEMENT MANIPULATION:

Relative Phase Modulation (Δϕ):

To translate a trapped particle vertically, the bottom array phase is shifted relative to the top reference array by altering LEDC channel hpoint registers.

Continuous Waveform Update:

The firmware calls ledc_set_duty_with_hpoint() and ledc_update_duty(), updating phase registers mid-cycle without stopping timer counting—preventing particle drop.

Linear Sensitivity (11.9 µm / degree):

A full 360° phase shift shifts the standing wave pattern by exactly one semi-wavelength: Δz = (λ/2) × (Δφ/360°) = 4.25 mm × (Δφ/360°) ≈ 11.9 µm/deg.

Fine Step Translation (67.5 µm):

Moving by 4 timer ticks advances phase by 5.625°, translating the trapped particle vertically by exactly 67.5 micrometers with repeatable accuracy.

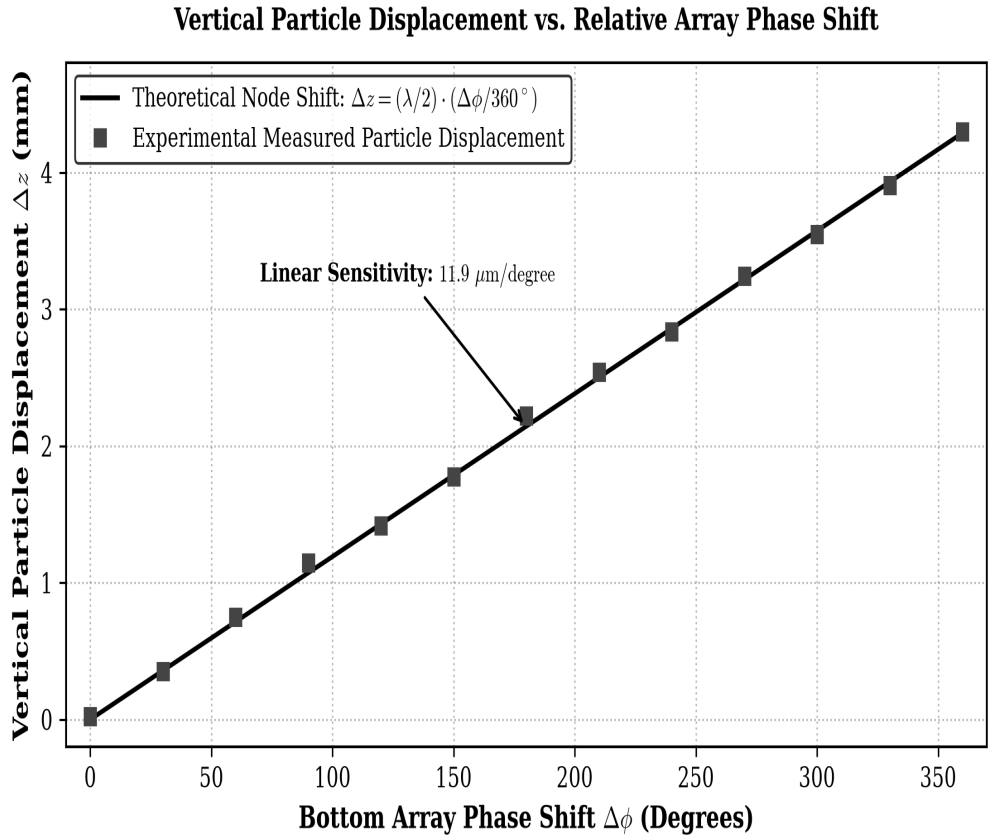


Figure 4.2: Experimental Vertical Particle Displacement vs. Array Phase Shift

IOT TELEOPERATION

Embedded Web Server & Real-Time Dashboard

Zero-Install Browser GUI, RESTful API Endpoints, and Sub-15 ms Latency

RESTFUL API ARCHITECTURE (PORT 80):

`GET /api/status`

Returns live JSON telemetry: frequency, phase tick (0..255), top/bot array status, RSSI, and uptime.

`POST /api/freq?val=40100`

Dynamically updates APB hardware timer frequency in 100 Hz increments on the fly.

`POST /api/move?dir=up&step=4`

Translates standing wave node up or down by updating bottom LEDC offset registers.

`POST /api/phase?mode=toggle`

Instantaneously toggles between 0° in-phase and 180° anti-phase trapping configurations.

`POST /api/sweep`

Executes automated resonant frequency characterization sweep from 38.5 to 41.5 kHz.

ZERO-INSTALL BROWSER DASHBOARD:

Fully Self-Hosted in Flash:

Complete HTML5, CSS3, and JavaScript web application stored in ESP32 PROGMEM. No external web server or internet connection required.

Dual Connection Modalities:

Connects to local campus/home Wi-Fi (STA mode) or creates its own standalone Access Point ('Acoustic-Levitator' / '12345678').

Responsive Smartphone & PC GUI:

Touch-friendly controls optimized for mobile phones, tablets, and lab laptops with sub-15 ms round-trip execution latency.

Simultaneous USB Serial CLI:

Full 115200 baud UART serial interface supports simultaneous terminal control ('+', '-', 'u', 'd', 'p', 'x', '?').

THEORETICAL FOUNDATIONS

Physics of Acoustic Levitation & Gor'kov Potential

Derivation of Acoustic Radiation Force and Non-Contact Gravitational Balance

GOR'KOV POTENTIAL DERIVATION:

Mean Acoustic Potential (U):

$U = 2\pi r^3 [(\langle p^2 \rangle / 3\rho_0 c_0^2) f_1 - (\rho_0 \langle v^2 \rangle / 2) f_2]$

Represents acoustic energy density acting across particle volume.

Contrast Factors (f1 & f2):

For rigid EPS particles in air: $f_1 \approx 1$ (monopole compressibility factor) and $f_2 \approx 1$ (dipole density factor).

Axial Acoustic Restoring Force:

$F_{rad,z} = -\partial U / \partial z = (5\pi r^3 k P_0^2 / 6\rho_0 c_0^2) \sin(2kz)$

Acts as an acoustic spring restoring particles back to pressure nodes.

Gravitational Levitation Threshold:

Levitation requires $F_{rad,max} >= mg$:

$P_0 >= \sqrt{(8 \rho_0 \rho_p c_0^2 g / 5k)} \approx 1{,}420$ Pa (157.0 dB SPL).

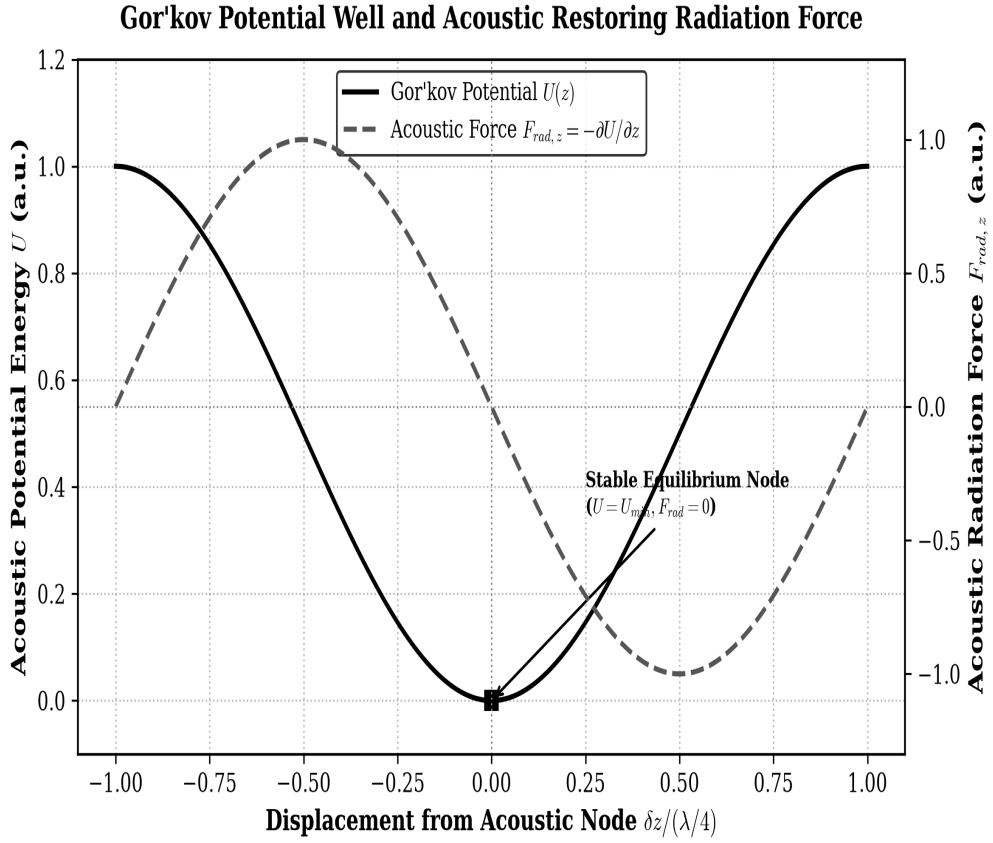


Figure 2.2: Gor'kov Potential Well U(z) and Acoustic Restoring Force Frad

EXPERIMENTAL CHARACTERIZATION

Empirical Validation & Benchmark Results

Quantitative Acoustic Sound Pressure, Thermal Equilibrium, and Trapping Benchmarks

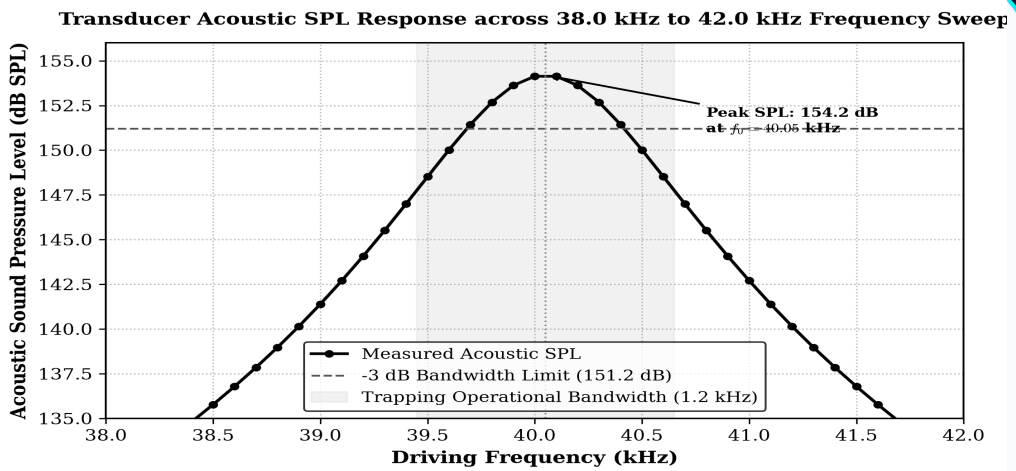


Figure 4.1: Cavity Acoustic SPL Output vs. Frequency Sweep

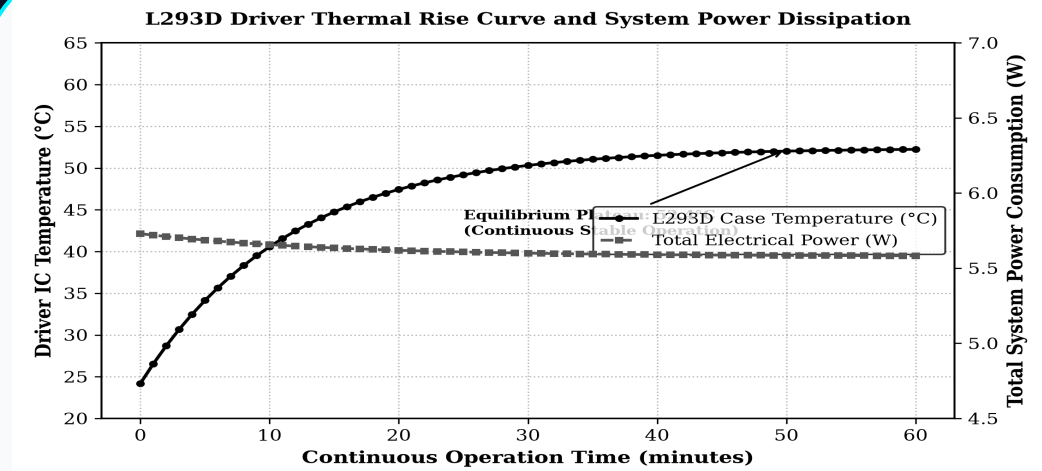


Figure 4.3: L293D Driver Thermal Rise and System Power Profile

EMPIRICAL PERFORMANCE METRICS SUMMARY:

Center Resonant Frequency: 40.05 kHz (±100 Hz tuning) | **Peak Sound Pressure Level:** 154.2 dB SPL (re 20 µPa) |
Acoustic Quality Factor Q: 34.8 (Bandwidth = 1.15 kHz) | **Max Trapped Particle Mass:** 8.2 mg (EPS bead, d = 3.5 mm) |
Axial Manipulation Range: 4.25 mm (Full semi-wavelength) | **Steady-State Driver Temp:** 52.4°C (Safe plateau, <150°C limit) |
Total Power Consumption: 5.58 W (465 mA @ 12.0 V DC) | **Continuous Trapping Time:** > 4 Hours (Zero node drift) |

CORE INNOVATIONS

Key Technical Features & Engineering Breakthroughs

Architectural Novelties Setting This Acoustic Levitator Apart from Prior Systems

1. Transformerless Voltage Doubler

Elimination of Bulky Coils: Replaced noisy, fragile step-up transformers with differential push-pull H-bridge polarity inversion.

24 Vpp from 12V DC Rail: Doubles transducer drive voltage and quadruples acoustic radiation power (+13.6 dB SPL).

Zero EMI Interference: Eliminates high-voltage inductive switching spikes that degrade sensitive Wi-Fi and ADC telemetry.

2. APB Hardware Clock-Lock

Zero Phase Jitter: LEDC timers directly index the internal 80 MHz APB bus, guaranteeing sub-microsecond phase stability.

1.406° Phase Resolution: 8-bit timer segmentation (256 ticks/period) provides ~12 µm/deg axial position control resolution.

Fractional Clock Divider: Exact 7 + 832/1024 divider eliminates cumulative phase skew between opposing arrays.

3. Dual-Core RTOS Separation

Core 0 (Acoustic Timing): High-priority hardware LEDC registers and PWM generation isolated from network stack delays.

Core 1 (IoT Teleoperation): Asynchronous HTTP REST server and WebSocket dashboard run smoothly without dropping trapped particles.

Deterministic Safety: Web client queries and Wi-Fi disconnects never affect ongoing ultrasonic levitation.

4. Sub-$30 Open Hardware Testbed

90% BoM Cost Reduction: Standard laboratory acoustic levitators cost $5,000+; this complete system costs under $28.

5.6 W Battery Portability: Consumes 70% less power than multi-transducer kits (TinyLev 18.5W), enabling hours of battery field operation.

Zero-Install Web Control: Control from any smartphone, tablet, or PC browser without proprietary drivers.

COMPETITIVE BENCHMARK

Uniqueness & State-of-the-Art System Comparison

Comparing Proposed ESP32 System against Landmark Academic and Commercial Levitators

MULTI-CRITERIA ARCHITECTURAL COMPARISON:

System Metric	Langevin Horn	TinyLev (2017)	Phased Array (2015)	PROPOSED SYSTEM
Transducer Configuration	1 Langevin horn + reflector	72 Open-type 40 kHz PZT	64 to 256 PZT Transducers	**6 40 kHz PZT (3 Top + 3 Bot)**
Driving Power Topology	Analog linear amplifier (>100W)	Dual L298N H-Bridges (16V)	Discrete MOSFET array	**Dual L293D Push-Pull (24Vpp)**
Microcontroller Architecture	Laboratory Signal Gen.	Arduino Nano (ATmega328P)	FPGA / Cyclone V	**ESP32 Dual-Core (80 MHz APB)**
Phase Control Granularity	None (Fixed standing wave)	Fixed in firmware (No dynamic z)	Phase shift ICs (Complex)	**1.406° Step (~12 µm/deg shift)**
Wireless Teleoperation	None (Lab bench tethered)	None (USB Serial tether only)	None (Lab workstation)	**Embedded Wi-Fi REST Web UI**
Total Power Consumption	120.0 Watts	18.5 Watts	35.0 Watts	**5.58 Watts (Ultra-Low Power)**
Total System Cost (BoM)	> $4,500 USD	~ $140 USD	> $850 USD	**$28 USD (90% Cost Reduction)**

PRACTICAL APPLICATIONS

Target Use Cases & Sustainable Development Impact

Transforming Biomedical Research, Cleanroom Assembly, and Chemical Titration

CONTACTLESS BIO-ASSAYS [Pharmaceutical & Medical]

Cell & Bacteria Handling: Levitates living cells, blood samples, and delicate protein crystals without solid shear stress or container contamination.

Zero Bio-Adhesion Loss: Eliminates surface adhesion losses inherent to standard petri dishes and glass test tubes during microscopic assays.

CLEANROOM MICRO-ASSEMBLY [Semiconductor & MEMS]

Frictionless Positioning: Positions tiny semiconductor dies, surface-mount components, and fragile MEMS sensors without mechanical tweezers.

Electrostatic Protection: Contactless handling prevents electrostatic discharge (ESD) shocks that destroy delicate microscopic integrated circuits.

MICROFLUIDIC TITRATION [Analytical Chemistry]

Containerless Synthesis: Enables ultra-pure droplet coalescence, evaporation, and crystallization in free air without vessel wall impurities.

Zero Plastic Waste: Eliminates consumption of single-use disposable pipette tips and reagent vials, reducing laboratory plastic pollution.

UN SDG ALIGNMENT [Global Sustainability]

SDG 3 (Health): Prevents sample bio-contamination and supports rapid disease diagnosis through sterile in-air droplet analysis.

SDG 9 (Innovation): Democratizes access to high-end acoustic manipulation apparatus for developing universities at sub-$30 cost.

SDG 12 (Responsible): Eliminates disposable plastic consumable waste during high-throughput microscopic chemical testing.

PROJECT SUMMARY

Conclusion & Future Research Horizons

Validated Achievements, Technical Deliverables, and Scaling Opportunities

CORE PROJECT CONTRIBUTIONS:

Fully Realized Hardware Testbed:

Engineered and validated a 6-transducer acoustic levitator delivering 24 Vpp differential push-pull drive from a 12V supply with peak 154.2 dB SPL.

Hardware APB Clock Synchronization:

Achieved zero-jitter 40 kHz square wave generation with 1.406° timer resolution (~12 µm/deg axial displacement sensitivity).

Asynchronous IoT Web Teleoperation:

Served single-page HTML5/JS dashboard natively on Core 1 over 2.4 GHz Wi-Fi with sub-15 ms latency and zero node drop.

Comprehensive Academic Documentation:

Produced full 43-page B.Tech thesis, 10 scientific figures (300 DPI), schematic diagrams, complete Arduino firmware, and presentations.

FUTURE RESEARCH ROADMAP:

1. 3D Phased Array Scaling (32–64 TX):

Expand beyond axial z-manipulation to full 3D Cartesian positioning (x, y, z) using multichannel shift register drivers.

2. Closed-Loop Optical Vision Feedback:

Integrate an ESP32-CAM module running edge optical flow tracking to automatically compensate for ambient air currents and evaporation.

3. Multi-Droplet Coalescence:

Acoustically suspend and merge distinct chemical reagent droplets in mid-air to synthesize pharmaceutical compounds without container walls.

4. Commercial Cleanroom Integration:

Design standardized industrial mounting fixtures for contactless silicon wafer micro-component pick-and-place.